EXACT HEAT-SHAPE DESENSITIZATION WHEN THE CONTROL CONTAINS THE OBSERVATION BOUNDARY

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Editorial note. Public 2109 is finite-dimensional exactness plus global approximation in disjoint geometry. The present target is all-direction exactness in a boundary-collar geometry, obtained by a separate cutoff/commutator construction; the interior-observation easy case is not counted.

Allowed standard prerequisites (editorial).

- Classical heat equation and functional analysis. Weak and transposition solutions, Sobolev trace maps and parabolic regularity, Holmgren unique continuation, Hilbert-space range/kernel duality, standard heat observability and convex minimization. The displayed Hadamard derivative formula is a standard supplied input. Target-specific constructions remain in the original proof.

Let $T > 0$ denote a horizon of time, ω and Θ be two open subsets of $\mathbb{R}^d$, $d \in \mathbb{N}^*$, and $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$.

For a bounded, connected and open set Ω of $\mathbb{R}^d$ of class $\mathscr{C}^2$, we consider the shape functional J_h, indexed by $h \in L^2(0,T;L^2(\omega))$, defined by

$$J_h(\Omega) = \frac{1}{2}\int_0^T \int_\Theta y_{\Omega,h}(t,x)^2 \, dx \, dt, \tag{1}$$

where $y_{\Omega,h} \in L^2((0,T)\times\mathbb{R}^d)$ is defined on $(0,T)\times\Omega$ as the unique weak solution of

$$\begin{cases} \dfrac{\partial y}{\partial t} - \Delta y &= \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega, \\ y &= 0 & \text{on } (0,T)\times\partial\Omega, \\ y(0,\cdot) &= 0 & \text{in } \Omega, \end{cases} \tag{2}$$

($\mathbb{1}_\omega$ denotes the characteristic function of the set ω) extended by 0 outside $(0,T)\times\Omega$.

In the sequel, given $\boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$, we introduce the family $\{\Omega_{\tau\boldsymbol{V}}\}_{\tau\in[0,\tau_{\boldsymbol{V}})}$ of domains defined by

$$\Omega_{\tau\boldsymbol{V}} = (\mathrm{Id} + \tau\boldsymbol{V})(\Omega_0).$$

Proposition 0.1 ([28, Proof of Prop. 1.1])**.** *Let $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$ and $h \in L^2(0,T;L^2(\omega))$. For all $\boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$, the mapping $\tau \mapsto J_h(\Omega_{\tau\boldsymbol{V}})$ is differentiable at $\tau = 0$ and*

$$\frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = \int_{\partial\Omega_0} \boldsymbol{V}\cdot\boldsymbol{n}\left(\int_0^T \partial_n y_0 \partial_n z_0 \, dt\right) d\sigma, \tag{3}$$

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where the pair (y_0, z_0) solves the coupled forward-backward system

$$
(4)\qquad \left\{\begin{array}{rcll} \dfrac{\partial y_0}{\partial t} - \Delta y_0 &=& \xi + h\mathbb{1}_\omega & \text{in } (0,T)\times\Omega_0,\\ y_0 &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ y_0\,(0,\cdot) &=& 0 & \text{in } \Omega_0,\\ -\dfrac{\partial z_0}{\partial t} - \Delta z_0 &=& \mathbb{1}_\Theta y_0 & \text{in } (0,T)\times\Omega_0,\\ z_0 &=& 0 & \text{on } (0,T)\times\partial\Omega_0,\\ z_0\,(T,\cdot) &=& 0 & \text{in } \Omega_0. \end{array}\right.
$$

($\mathbb{1}_\Theta$ denotes the characteristic function of the set Θ.)

Definition 0.2. Let $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$. We say that the control function $h \in L^2(0,T;L^2(\omega))$ desensitizes J_h exactly at Ω_0 at the first order with respect to domain variations if

$$
(5)\qquad \forall\, \boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d), \quad \frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = 0.
$$

Let $\mathcal{E}$ be a linear subspace of $W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d)$. We say that the control function $h \in L^2(0,T;L^2(\omega))$ exactly desensitizes J_h for $\mathcal{E}$ at Ω_0 at the first order with respect to domain variations if

$$
(6)\qquad \forall\, \boldsymbol{V} \in \mathcal{E}, \quad \frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0} = 0.
$$

Given $\varepsilon > 0$, we say that the control function $h \in L^2(0,T;L^2(\omega))$ ε-approximately desensitizes J_h at Ω_0 at the first order with respect to domain variations if

$$
(7)\qquad \forall\, \boldsymbol{V} \in W^{3,\infty}(\mathbb{R}^d,\mathbb{R}^d), \quad \left|\frac{d}{d\tau}\left(J_h\left(\Omega_{\tau\boldsymbol{V}}\right)\right)\Big|_{\tau=0}\right| \leq \varepsilon \|\boldsymbol{V}\|_{W^{3,\infty}(\mathbb{R}^d;\mathbb{R}^d)}.
$$

Theorem 0.3. *Let ω and Θ be smooth non-empty open subsets of Ω_0 such that*

$$
(8)\qquad \partial\Theta \text{ has only one connected component,} \quad \Theta \Subset \Omega_0, \quad \text{and} \quad \partial\Theta \subset \omega.
$$

Then, for all $\xi \in L^2(0,T;L^2(\Omega_0))$, there exists $h \in L^2(0,T;L^2(\omega))$ such that the functional J_h in (1) is exactly desensitized in the sense of (5).

0.1. **Proof of Theorem 0.3.** We start by introducing open sets ω_0, ω_1, ω_2, and ω_3 such that

$$
\partial\Theta \subset \omega_0 \Subset \omega_1 \Subset \omega_2 \Subset \omega_3 \Subset \omega,
$$

which is possible thanks to Assumption (8). We also introduce a smooth function $\eta_{23} = \eta_{23}(x)$ taking value 1 in $\Omega \setminus \overline{\omega_3}$ and vanishing in $\overline{\omega_2}$.

We fix $\xi \in L^2(0,T;L^2(\mathbb{R}^d))$, and we introduce the solution y_ξ to

$$
\left\{\begin{array}{rcll} \partial_t y_\xi - \Delta y_\xi &=& \eta_{23}\xi, & (t,x) \in (0,T)\times\Omega_0,\\ y_\xi(t,x) &=& 0, & (t,x) \in (0,T)\times\partial\Omega_0,\\ y_\xi(0,x) &=& 0, & x \in \Omega, \end{array}\right.
$$

and the solution z_ξ to

$$
(9)\qquad \left\{\begin{array}{rcll} -\partial_t z_\xi - \Delta z_\xi &=& \eta_{12} y_\xi \mathbb{1}_\Theta, & (t,x) \in (0,T)\times\Omega_0,\\ z_\xi(t,x) &=& 0, & (t,x) \in (0,T)\times\partial\Omega_0,\\ z_\xi(T,x) &=& 0, & x \in \Omega_0. \end{array}\right.
$$

where $\eta_{12} = \eta_{12}(x)$ is a smooth function taking value 1 in $\Omega_0 \setminus \overline{\omega_2}$ and vanishing in $\overline{\omega_1}$.

Then, we introduce a smooth function $\eta_{01} = \eta_{01}(x)$ such that η_{01} vanishes in ω_0 and equal to 1 in $\Omega_0 \setminus \omega_1$, so that $\mathbb{1}_\Theta \eta_{01}$ is actually a smooth function taking value

1 in $\Theta \setminus \omega_1$ and vanishing in $\omega_0 \cup (\Omega_0 \setminus \Theta)$. Then $z_0(t,x) = \mathbb{1}_\Theta(x)\eta_{01}(x)z_\xi(t,x)$ satisfies

$$\begin{cases} -\partial_t z_0 - \Delta z_0 &= \eta_{01}\eta_{12}y_\xi \mathbb{1}_\Theta - [\Delta, \mathbb{1}_\Theta\eta_{01}]z_\xi, & (t,x) \in (0,T)\times\Omega_0, \\ z_0(t,x) &= 0, & (t,x) \in (0,T)\times\partial\Omega_0, \\ z_0(T,x) &= 0, & x \in \Omega_0, \end{cases}$$

and, by construction,

$$z_0(t,x) = 0 \quad \text{for all } (t,x) \in (0,T)\times(\Omega_0 \setminus \Theta).$$

Now, we remark that by construction, $[\Delta, \mathbb{1}_\Theta\eta_{01}]z_\xi$ is localized in $\omega_1 \cap \Theta$. Besides, local regularity results for (9) imply that z_ξ is $\mathscr{C}^2([0,T]\times\overline{\omega_1\cap\Theta})$. We then take

$$y_0(t,x) = \eta_{01}(x)\eta_{12}(x)y_\xi(t,x) - [\Delta, \mathbb{1}_\Theta\eta_{01}]z_\xi(t,x), \quad \text{for all } (t,x) \in (0,T)\times\Omega_0,$$

which satisfies (4) for h given for $(t,x) \in (0,T)\times\Omega_0$ by

$$h(t,x) = (\eta_{01}(x)\eta_{12}(x)\eta_{23}(x) - 1)\xi(t,x) - [\Delta, \eta_{01}\eta_{12}]y_\xi(t,x) \\ - (\partial_t - \Delta)([\Delta, \mathbb{1}_\Theta\eta_{01}]z_\xi)(t,x).$$

This control function h is localized in $(0,T)\times\omega$ due to the conditions on the support of η_{01}, η_{12}, η_{23}. This concludes the proof of Theorem 0.3.

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